

# OUEDRAOGO Sayouba

Network and Systems Infrastructure Engineering

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## PROFILE

Engineering student in Network Engineering and Information Systems with hands-on experience in network design, infrastructure deployment, systems administration and cybersecurity. Solid foundation in TCP/IP protocols, Cisco configuration, Linux server hardening and network simulation (Cisco Packet Tracer, OMNeT++, Wireshark). Active CTF competitor with multiple top rankings in national cybersecurity challenges, combining technical rigor with a problem-solving mindset.

## TECHNICAL SKILLS

|                        |                                                                                      |
|------------------------|--------------------------------------------------------------------------------------|
| <b>Networking</b>      | TCP/IP, Subnetting, VLANs, Routing (OSPF, RIP), Switching, Spanning Tree             |
| <b>Network Tools</b>   | Cisco Packet Tracer, OMNeT++, Wireshark, Nmap                                        |
| <b>Security</b>        | Firewall config (Fortinet), IDS/IPS basics, vulnerability assessment, CTF techniques |
| <b>Systems Admin</b>   | Linux (Debian/Ubuntu), Windows Server 2019, Active Directory basics                  |
| <b>Scripting / Dev</b> | Python (automation), Bash, Java, Git/GitHub                                          |
| <b>Virtualisation</b>  | VirtualBox, VMware basics, Docker basics                                             |

## PROJECTS

### Enterprise Network Design & Simulation

Designed and simulated a multi-site enterprise network with VLAN segmentation, inter-VLAN routing, OSPF dynamic routing and redundancy mechanisms. Analyzed packet behavior and link performance under varying traffic loads.

*Cisco Packet Tracer · OMNeT++ · TCP/IP · OSPF · STP*

**Role:** Full design — topology planning, IP addressing scheme, routing configuration and performance analysis.

### Secure Infrastructure Lab — Windows Server 2019

Set up a small enterprise infrastructure with Active Directory, DNS and DHCP services. Configured user roles, GPO-based access control policies and mapped shared network resources with permission tiers.

*Windows Server 2019 · Active Directory · DNS · DHCP · GPO*

**Role:** Solo lab — domain controller setup, user and group policy management, service configuration.

### Network Monitoring & Traffic Analysis

Deployed a network monitoring environment to capture and analyze live traffic, identify anomalies and interpret protocol-level behavior. Studied ARP, ICMP, TCP handshake and common attack signatures.

*Wireshark · Nmap · Linux · Bash*

**Role:** Full setup — packet capture, filter scripting and protocol analysis report.

### Linux Server Hardening & Service Deployment

Configured and hardened a Linux server: SSH key-based authentication, UFW firewall rules, fail2ban, automated log monitoring and deployment of web and FTP services.

*Linux (Debian) · Bash · UFW · fail2ban · Apache · vsftpd*

**Role:** Solo project — system hardening, service deployment and automated monitoring scripts.

### Home Automation Network (IoT Simulation)

Simulated a smart home network integrating IoT devices with a central gateway, applying QoS policies, NAT and access control lists to manage device traffic and ensure network security.

*Cisco Packet Tracer · IoT module · ACL · NAT · QoS*

**Role:** Full design — device topology, traffic prioritisation rules and security policy configuration.

## EDUCATION

|                |                                                                                |
|----------------|--------------------------------------------------------------------------------|
| 2024 – Present | <b>Engineering Degree – Networks &amp; Information Systems</b> · FST Marrakech |
| 2022 – 2024    | Diploma of Studies in Sciences and Techniques · FST Settat                     |
| 2019 – 2022    | Technical Baccalaureate (Mathematics, Physical Sciences) · LSN, Bobo Dioulasso |

## HACKATHONS & COMPETITIONS

|                                             |                                                                   |
|---------------------------------------------|-------------------------------------------------------------------|
| <b>HackShield — SecDojo &amp; AXA GBS</b>   | Top 10 / 200 teams — securing simulated IT infrastructures        |
| <b>AUSIM CyberDrill — East Region V2</b>    | Top 20 / 50 teams — CVE exploitation & RCE, 18h                   |
| <b>GCDxJIT CTF</b>                          | Top 5 / 30 teams — Forensics, DFIR, PWN, 15h                      |
| <b>CyberOdyssey CTF — SecDojo</b>           | 57 <sup>th</sup> / 414 teams — 24h cybersecurity competition      |
| <b>GCDxN7 CTF — Code 212 Marrakech</b>      | 15 <sup>th</sup> / 40 teams — network security & traffic analysis |
| <b>Orange Digital Center Champions 2025</b> | Top 5 / 50 national — algorithmic programming                     |

## CERTIFICATIONS & LANGUAGES

**Cisco:** Networking Basics · Equipment & Basic Configuration · Network Administration & Troubleshooting · Linux & Windows OS Basics

**Fortinet:** FCF – Introduction to Cybersecurity · Technical Introduction to Cybersecurity · Threat Landscape

**Languages:** French (C1 – Fluent) | English (B1 – Intermediate)